Akı Kuantumlanması

$$H=\frac{1}{2\mu}\left(p+\frac{e}{c}A\right)^2+V \Rightarrow H\psi=E\psi$$

$$\frac{1}{2\mu}\left(p+\frac{e}{c}A\right)\left(p+\frac{e}{c}A\right)\psi+V\psi=E\psi \rightarrow \frac{1}{2\mu}\underbrace{\left(\frac{\hbar}{i}\nabla+\frac{e}{c}A\right)\left(\frac{\hbar}{i}\nabla+\frac{e}{c}A\right)}\psi_0 e^{-i\left(\frac{e}{\hbar c}\right)f}+V\psi_0 e^{-i\left(\frac{e}{\hbar c}\right)f}=E\psi_0 e^{-i\left(\frac{e}{\hbar c}\right)f}$$

$$\left(\frac{\hbar}{i}\nabla+\frac{e}{c}A\right)\left(\frac{\hbar}{i}\underbrace{\nabla\left(e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0\right)}_{(i)}+\frac{e}{c}A\left(e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0\right)\right)$$

$$(i)\ \frac{\hbar}{i}\cdot\left(\frac{-ie}{\hbar c}\right)\nabla f\, e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0+\frac{\hbar}{i}e^{-i\left(\frac{e}{\hbar c}\right)f}\nabla\psi_0$$

$$\left(\frac{\hbar}{i}\nabla+\frac{e}{c}A\right)\left(\left(-\frac{e}{c}\right)\nabla f\, e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0+\frac{\hbar}{i}e^{-i\left(\frac{e}{\hbar c}\right)f}\nabla\psi_0+\frac{e}{c}A\left(e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0\right)\right)$$

$$\left(\underbrace{\frac{\hbar}{i}\nabla}_{1}+\underbrace{\frac{e}{c}A}_{2}\right)\left(\frac{e}{c}\underbrace{(A-\nabla f)}_{a}e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0+\underbrace{\frac{\hbar}{i}e^{-i\left(\frac{e}{\hbar c}\right)f}}_{b}\nabla\psi_0\right)$$

$$(1,a)\ \frac{\hbar}{i}\frac{e}{c}(\nabla A-\nabla^2 f)e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0+\frac{\hbar}{i}\frac{e}{c}(A-\nabla f)\left(-\frac{ie}{\hbar c}\right)\nabla f\, e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0+\frac{\hbar}{i}\frac{e}{c}(A-\nabla f)e^{-i\left(\frac{e}{\hbar c}\right)f}\nabla\psi_0$$

$$(1,b)\ \frac{\hbar}{i}\left(\frac{\hbar}{i}\right)\left(-\frac{ie}{\hbar c}\right)\nabla f\, e^{-i\left(\frac{e}{\hbar c}\right)f}\nabla\psi_0+\frac{\hbar}{i}\frac{\hbar}{i}e^{-i\left(\frac{e}{\hbar c}\right)f}\nabla^2\psi_0$$

$$(2,a)\ \frac{e^2}{c^2}(A^2-A\nabla f)e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0$$

$$(2,b)\ \frac{\hbar}{i}\frac{e}{c}e^{-i\left(\frac{e}{\hbar c}\right)f}A\nabla\psi_0$$

$$e^{-i\left(\frac{e}{\hbar c}\right)}\left\{\frac{\hbar}{i}\frac{e}{c}\left\{(\nabla A-\nabla^2 f)\psi_0-\frac{e}{c}(A-\nabla f)(\nabla f)\psi_0+(A-\nabla f)\nabla\psi_0-(\nabla f)\nabla\psi_0+\frac{\hbar}{i}\frac{c}{e}\nabla^2\psi_0+\frac{i}{\hbar}\frac{e}{c}(A^2-A\nabla f)\psi_0+A\nabla\psi_0\right\}\right\}$$

$$\frac{1}{2\mu}\left\{\left(p+\frac{e}{c}A\right)^2\psi_0\right\}\Rightarrow \psi=e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0 \Rightarrow \psi_0=e^{i\left(\frac{e}{\hbar c}\right)f}\psi$$

$$\Rightarrow\frac{1}{2\mu}\left\{\left(\frac{\hbar}{i}\nabla+\frac{e}{c}A\right)\left(\frac{\hbar}{i}\underbrace{\nabla e^{i\left(\frac{e}{\hbar c}\right)f}\psi}_{*}+\frac{e}{c}A\, e^{i\left(\frac{e}{\hbar c}\right)f}\psi\right)\right. \qquad \psi=e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0$$

$$\Rightarrow *\quad \nabla\psi_0=\nabla\left(e^{i\left(\frac{e}{\hbar c}\right)f}\psi\right)=\left(i\frac{e}{\hbar c}\right)(\nabla f)e^{i\left(\frac{e}{\hbar c}\right)f}\psi+e^{i\left(\frac{e}{\hbar c}\right)f}\nabla\psi$$

$$\nabla^2\psi_0=\nabla\cdot(\nabla\psi_0)=\nabla\cdot\left(\left(\frac{ie}{\hbar c}\right)\nabla f\, e^{\frac{ie}{\hbar c}f}\psi+e^{\frac{ie}{\hbar c}f}\nabla\psi\right)$$

$$\Rightarrow\frac{ie}{\hbar c}\left\{(\nabla^2 f)e^{i\left(\frac{e}{\hbar c}\right)f}\psi+\left(\frac{ie}{\hbar c}\right)(\nabla f)^2 e^{\left(\frac{ie}{\hbar c}\right)f}\psi+\nabla f\, e^{\left(\frac{ie}{\hbar c}\right)f}\nabla\psi+\nabla f\, e^{\left(\frac{ie}{\hbar c}\right)f}\nabla\psi\right\}+e^{\left(\frac{ie}{\hbar c}\right)f}\nabla^2\psi$$

$$e^{(ie/\hbar c)f}\left\{\left(\frac{ie}{\hbar c}\right)\left\{(\nabla^2 f)\psi+(\nabla f)^2\psi+(\nabla f)(\nabla\psi)+\nabla f\nabla\psi\right\}+\nabla^2\psi\right\}$$

$$\left((\nabla f)^2+2\nabla f+\nabla^2 f\right)$$

$$\left(\frac{\hbar}{i}\nabla+\frac{e}{c}\nabla f\right)^2 \qquad \frac{\hbar}{i}+\frac{e}{c}A \qquad \boxed{A=\nabla f}$$

$$e^{i\left(\frac{e}{\hbar c}\right)f}\left\{\frac{\hbar e}{i c}\left\{\underbrace{(\nabla A-\nabla^2 f)}_{0}\psi_0-\frac{e}{c}\underbrace{(A-\nabla f)}_{0}(\nabla f)\psi_0+\underbrace{(A-\nabla f)}_{0}\nabla\psi_0-(\nabla f)\nabla\psi_0+\frac{\hbar}{i}\frac{c}{e}\nabla^2\psi_0+\frac{i e}{\hbar c}\underbrace{(A^2-A\nabla f)}_{0}\psi_0+A\nabla\psi_0\right\}\right\}\quad\boxed{A=\nabla f}$$

$$\frac{1}{2\mu}\left\{e^{i\left(\frac{e}{\hbar c}\right)f}\left\{\frac{\hbar}{i}\frac{e}{c}\right\}\left(A(\nabla\psi_0)-\underbrace{(\nabla f)}_{A}\nabla\psi_0+\frac{\hbar}{i}\frac{c}{e}\nabla^2\psi_0\right)\right\}$$

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi_0+V(r)\psi_0=E_0\quad\boxed{A=\nabla f}$$

$$A=\nabla f\Rightarrow f(r)\Rightarrow\int dr\,A\qquad\psi(r)=e^{-i\left(\frac{e}{\hbar c}\right)f}\psi_0$$

$$f(r)=\int dr'\,A(r')\quad=e^{-i\left(\frac{e}{\hbar c}\right)f}\int dr\,A$$

$$\psi(r)=e^{-i\left(\frac{e}{\hbar c}\right)\int dr'\,A(r')}\psi_0$$

Stokes Teoremi

$$\oint dr\,A(r)=\int \underbrace{ds'\,(\nabla\times A)}_{\int ds\,B=\Phi}$$